



Doubao I

The Doubao Agent System: Five-Layer Architecture, Form Genealogy and Production SOP from Model Substrate to OS Agent

豆包智能体体系：五层架构与生产 SOP



The Doubao Agent System: Five-Layer Architecture, Form Genealogy and Production SOP from Model Substrate to OS Agent

One-sentence conclusion: in the two-plus years from 2024-05 to 2026-09, Doubao layer by layer encapsulated "large model capability" into a **L1 model substrate → L2 protocols → L3 development and enterprise platforms → L4 product layer → L5 C-end form** five-layer vertically integrated system, and with the three products "Doubao Work / Work Companion / phone assistant" occupied respectively the three autonomy tiers personal, team, system-level; but within the system the multi-Agent division mechanism, concurrency and context quotas, skill governance details still have 14 "to verify" items, success rate and function point numbers mostly official or media caliber, not independently reproducible.

Abstract

This paper, based on ByteDance/Volcano Engine/Doubao official release materials, authoritative media reports and one internal architecture design specification sheet (containing three-state source marking), does one **structural anatomy** of the "Doubao Agent system" as of 2026-09-21. The full text does no vendor endorsement, only three things: (1) reorganizing products scattered across press conferences, releases, help centers into one comparable chart per the five-layer vertically integrated architecture; (2) separating the three conflated words "intelligent agent / Agent / super Agent," overlaying the L0–L5 autonomy ladder, marking Doubao each form's current tier; (3) organizing the spec sheet's ten component field categories, eight-stage SOP, four security gates into directly copyable PRD templates and acceptance checklists.

Evidence caliber note: the full text follows the spec sheet's three-state iron rule—[official statement] means official releases/press conferences/official doc originals; [media says] means authoritative media or press conference retellings; [to verify] means no official original obtained, body must not write as fact. UDOS side numbers (used in lower paper comparison) additionally carry verified / cpu-proto / unverified three levels.

Structured Abstract

- **Background and problem:** from 2025 second half, the four words "intelligent agent," "Agent," "super Agent," "OS Agent" conflated in Chinese tech media; Doubao, Coze, Feishu aily, TRAE, ArkClaw etc. products in 2026 underwent multiple organizational and brand integrations (2026-07-30 Feishu×Doubao organizational integration, 2026-08-24 Coze/TRAE merged into Doubao system), outsiders hard to see in one chart "who is at which layer, who serves whom, who calls whom."
- **Argument:** the Doubao system is not "a pile of Apps," but one **bottom-up reusable tech stack**—model substrate provides capability, protocol layer specifies interfaces, platform layer provides orchestration, product layer encapsulates scenarios, C-end form reaches users; five layers have clear dependency direction, Feishu side-hung as "context source."
- **Evidence:** five-layer architecture total 27 fact items (L1 models 10 + L2 protocols 4 + L3 platforms 7 + L4 products 3 + L5 C-end 3), six-form×nine-dimension matrix 54 cells, SOP 8 stages×4 roles+4 gates, ten component categories about 60 field rows; this independent online spot check 14 key facts, fully confirmed 11, partially confirmed 2, containing caliber corrections 2, **not one falsified**.
- **Contribution:** (1) one five-layer vertically integrated architecture chart and per-layer responsibility table; (2) three-word distinction+L0–L5 ladder+Agent-as-Tool vs Handoff graphic distinction; (3) six-form×nine-dimension native master table+heatmap overview; (4) eight-stage four-role swimlane chart+human review/red team/permission/canary four gates explicitly marked; (5) ten component field tables+four-question model selection decision tree.

1 Introduction: Why Redraw This Chart

2024-05-15 when Doubao large model first released, outsiders only saw "yet another domestic large model." Two years later, Doubao no longer a single App: below it hangs self-developed model families (2.0 / 2.1 Pro/Turbo / 1.5 deep thinking / UI-TARS / Seedance), in the middle run MCP/A2A/SAEP/Feishu CLI four protocol sets, middle layer has Coze/TRAE/AgentKit/HiAgent 3.0/ArkClaw/AI Trust six development and enterprise platforms, above has Doubao Work/Work Companion/phone assistant

three products, at the very top also Doubao App Agent plaza, Feishu 8.0, partner vendor OS entries three C-end forms.

The question: **are these five layers placed parallel or with dependency direction? Are "intelligent agent," "Agent," "super Agent" actually one thing? An enterprise genuinely wanting to deploy an Agent for its team, starting from which step, stopping at which step, who signs?** This paper tries to answer these three questions clearly with one chart, one table, one SOP set.

Methodology statement: this paper's all product names only describe public positioning, not endorsing vendors; all numbers and dates per spec sheet original values, online verification found caliber deviations separately marked, not rewriting spec sheet original values; not fabricating any DOI, concurrency count, context length.

Chapter summary: the Doubao system in 2026 already evolved from "one chat App" to "five-layer tech stack + three C-end forms," but public materials scattered across dozens of releases, needing one structural reorganization.

2 Five-Layer Vertically Integrated Architecture (Deliverable ①)



分层为作者基于公开资料归纳; 【媒体称】项未取得官方原文, 不提升为事实

示意 · 非官方架构图 · 2026-09-21

Five layers bottom-up: **L1 model substrate** → **L2 protocols** → **L3 development and enterprise platforms** → **L4 product layer** → **L5 C-end form**. Inter-layer arrows bottom-up, meaning "lower layer provides capability to upper layer." Feishu side-hung on the right as "work platform/context source," not an independent sixth layer.

2.1 L1 Model Substrate Layer: Self-Developed Model Family + GUI Operation Model

L1 is the whole system's capability origin. It is not a single model, but a **model family split by tier and modality**.

Model / version	Date	Positioning	Three-state
Doubao large model first release (originally Skylark)	2024-05-15	ByteDance self-developed multimodal family first release, pro/lite general + voice/text-to-image total 9; Volcano Engine原动力 conference	official
Doubao-1.5-pro	2025-01-22	Brand new base model, declares not using other models' distillation data	official
Doubao 1.5 deep thinking + UI-TARS-1.5 open source	2025-04-17	Deep thinking/multimodal reasoning enhanced; UI-TARS-1.5 SOTA on 7 GUI benchmarks; same day pushes OS Agent plan	official/paper
Seedance 1.0 lite	2025-05-13	Video generation model; same day Doubao 1.5 visual deep thinking	official
Doubao 1.8	2025-12-18	Multimodal understanding/generation/Agent capability into global first tier; daily tokens broke 5 trillion	official
Doubao 2.0 / Doubao-Seed-2.0	2026-02-14	Pro/Lite/Mini three general Agent models + Code model; 2.0 Pro on Doubao "expert" mode	official
Doubao 2.1 Pro / Turbo	2026-06-24	Agent model for real-world productivity; Pro connected to pro version, Turbo for free quota experience; API side Turbo price about half Pro	official
UI-TARS (open-source version)	2025-01 (arXiv: 2501.12326)	Native GUI Agent, ByteDance Seed × Tsinghua; github.com/bytedance/UI-TARS	official/paper
UI-TARS (closed-source version)	2025-12 (with tech preview)	Doubao phone assistant GUI operation substrate, optimized for Mobile Use, performance better than open-source version	media says
UI-TARS 2.0	date to verify	Points out pure GUI limits, connects external file systems and sandbox platforms via SDK	media says/to verify

API billing reference: 2.1 Pro input 6 yuan / output 30 yuan per million Tokens; Turbo about half. [spec sheet original value, official not public concurrency/context length details]

2.2 L2 Protocol Layer: Connecting Tools, Data Sources, Cross-Agent

L2 solves "how the model connects to the external world." The Doubao system simultaneously took four protocol paths:

Protocol	Full name / positioning	Role in Doubao system	Three-state
MCP	Model Context Protocol	Phone assistant second-generation integration, for AI connecting tools and data sources; only integrated under premise application side actively opens MCP service and authorizes	media says
A2A	Agent-to-Agent	Second-generation phone system AI and App built-in agents (Meituan/Alipay etc.) directly interoperate at bottom layer	media says
SAEP	Screen Automation Execution Protocol, screen automation operation declaration	Launched by Doubao, from 2026-09-14 entering 30-day public notice period ; third-party apps autonomously declare allowing/refusing/limiting AI automated operations within this app; during notice period not consented means not operate, after period not refused means default open; with traceable operation logs	media says (protocol launched by Doubao, notice rules per 36Kr/Sohu/Fengmian News)
Feishu CLI	Feishu open command line interface (open sourced, MIT, 12 business domains)	From 2026 end-March open to Agents; function points 247 → 767; call success rate 78% → 95%; call speed overall +39%	media says (Xie Xin 2026-09-15 speech)

SAEP's uniqueness: while MCP/A2A both solve "how Agents connect," SAEP first wrote "what application side allows Agents to do on its interface" into a **declarative boundary protocol**. This is Doubao's first creation at the protocol layer, but the 30-day notice period not yet ended, third-party apps' actual integration rate and disputes still to observe.

2.3 L3 Development and Enterprise Platform Layer: From Low Code to High Code

L3 is the "Agent-making factory." It covers from low-code orchestration to high-code development, from C-end publishing to enterprise privatization full spectrum.

Product	Positioning	Version / date	Three-state
Coze	Agent orchestration (low code), after orchestration publishable to Doubao/API	From 2026-08-24 overall merged into Doubao system	official/media
TRAE	Programming IDE / CLI / Work; TRAE Work, Coze and Doubao office capability integrated, TRAE IDE/CLI retained as programming product line	2026-08-24 merged	official/media (ByteDance response confirmed)
AgentKit	Enterprise Agent development platform	2026-06-24 upgraded	official
HiAgent 3.0	Enterprise Agent platform	2026-06-24	official
ArkClaw	Cloud SaaS OpenClaw, out-of-box, 7×24, dedicated ECS; with Volcano Coding Plan Pro	2026-03-09 launched (encyclopedia caliber; Volcano official version records first Ark-26.3.30 / 03-31)	official (existence) / media says (3-09 date)
AI Trust	Enterprise Agent security system	2026-06-24 released together with ArkClaw enterprise workbench	official
Ark CLI	Volcano Ark command line access	2026-06-24 released	official

2.4 L4 Product Layer: Divided by Service Object and Execution Environment

L4 is the "Agent productization" layer—the same L1 model substrate encapsulated into three products facing different users, running in different execution environments.

Product	Level positioning	Key facts	Three-state
Doubao Work	Personal level (personal productivity Agent)	2026-08-25 formally released independent client; local+cloud PC dual mode, Windows virtual desktop, phone remote controlling computer; skill center over 200 skills/connectors; browser cross-software operation; delivers docs/sheets/PPT/webpages/images/videos; Doubao App / Doubao Work App / Feishu three entries data interoperable	official
Doubao Work Companion	Team level (team Agent)	Domestic first team Agent , originally Feishu aily renamed/upgraded; independent identity/permissions/memory, enterprise unified configuration; Feishu CLI 767 function points; team long-term memory "one person teaches, whole team can use"	official statement
Doubao phone assistant	System level (OS Agent)	Cooperates with phone vendors at OS layer; ByteDance explicitly no self-developed phone plan ; UI-TARS closed-source substrate; system-level screenshot+event injection; sensitive operations (publish/delete/pay/logout) need human review or secondary confirmation	official statement

Work Companion rename date correction: Feishu aily official help center states "**from 2026 August 14, Feishu aily renamed 'Doubao Work Companion'**," from 2026-09-01 adds "proactive work" and bills; **2026-09-15** Feishu Future Infinite conference publicly released. Spec sheet records 2026-09-15 as release date, this paper records both dates.

Terminal pricing (public report caliber): Doubao pro version three tiers **68 / 200 / 500 yuan/month** (standard 68 connects 2.1 Pro, enhanced 200 = standard 4× quota, premium 500 = standard 10× quota;

standard quota over 5× free version). First-generation nubia M153 **3499 yuan, stocked 30,000, first batch sold out**; second-generation Nubia NaviX Ultra **12GB+512GB official price 5999 yuan, after national subsidy 5499 starting**.

2.5 L5 C-End Form Layer: Consumable/Creatable Roles Facing Users

L5 is the layer users genuinely "see."

Form	Positioning	Three-state
Doubao App "Agent" plaza	UGC interactive roles, creatable and consumable; public/private visibility	media says
Feishu 8.0	Systematically reconstructed for Agents, fully opens data and tools to Agents; in group chats search name to pull Agent into group	official (press conference)
Partner vendor OS entry	System layer; voice/side key/dedicated AI key/headset wake	official/media

2.6 Feishu Side-Hung Positioning (Separately Explained)

Feishu in this five-layer chart not the sixth layer, but **side-hung context source**: Doubao Work logs in via Feishu account, inheriting enterprise knowledge and work context within **permission scope** (group chat records, docs, meeting minutes, schedules), created content sediments back to Feishu; Doubao Work Companion natively runs within Feishu.

Feishu CEO Xie Xin says "Feishu no longer only serves people, must also start serving Agents"—[media says / press conference retelling], not written official document original.

2.7 AI Cloud-Native Architecture

L3 enterprise platform's bottom is the "AI cloud-native architecture": GPU as core compute unit, models exposed as services to upper layers; Ark CLI, AgentKit, HiAgent 3.0, ArkClaw all run above this layer. This layer's specific GPU specs, single-machine concurrency, cold-start latency etc. engineering parameters, official not public details.

Chapter summary: the five-layer architecture not marketing talk, but the real dependency chain "lower layer provides capability to upper layer"—L1 produces models, L2 produces interfaces, L3 produces tools, L4 produces products, L5 produces user touchpoints; Feishu side-hung as "in-permission context," not an independent sixth layer.

3 Concept Relationships: Three-Word Distinction, Nested Containment and Autonomy Ladder (Deliverable ②)



3.1 Three-Word Distinction: Intelligent Agent / Agent / Super Agent

Chinese tech circles in 2025–2026 used the three words almost as synonyms, but in the Doubao system they have clear tier differences:

- Intelligent agent (role type):** persona+knowledge base+plugins, conversational interaction, single-role memory. Landing Doubao App Agent plaza/UGC [media says].
- Agent (task autonomous type):** planning→tools→execution→delivery, cross-software, can split subtasks. Landing Doubao Work / Doubao Work Companion [official statement].
- Super Agent: not official product name**, media/industry umbrella term for "system-level OS Agent or multi-Agent organization." On the Doubao system corresponding to Doubao phone assistant, and Work Companion/work squad [media says].

This must be emphasized: "super intelligent agent/super Agent" in Doubao official materials never appeared as a product name. This paper follows the "media concept" label, not writing it as Doubao's official product line.

3.2 Nested Containment Relationship

The three words not parallel, but **nested containment**:

That is: one work Agent internally can contain multiple sub-Agents working in parallel; one OS Agent can simultaneously orchestrate multiple work/team Agents.

3.3 L0–L5 Autonomy Ladder (modeled on SAE J3016)

Borrowing autonomous driving SAE J3016's idea, dividing Agent autonomy also into six levels, marking Doubao each form's current tier:

Level	Meaning	Doubao form landing
L0 tool call	Call a tool once, return result	—
L1 Q&A	Single-round knowledge Q&A	UGC plaza L1–L2
L2 single task	Single-step tools complete one thing	Coze orchestration L2–L3
L3 long-process autonomous	Planning-execution-delivery	Doubao Work L3
L4 multi-Agent collaboration	Team formation division	Work Companion L3–L4
L5 system-level autonomous	OS-level cross-App • not achieved	Phone assistant • target L5 (current tech preview)

L5 not achieved: Doubao phone assistant in official caliber still "tech preview," second generation the consumer version; cross-App free operation reliability, long-tail App compatibility, user trust in "what AI clicked itself," all still climbing. This paper not writing it as "L5 achieved."

3.4 Agent-as-Tool vs Handoff: Two Multi-Agent Collaboration Modes

These two concepts often conflated in Chinese materials, but engineering-wise two things:

- **Agent-as-Tool (control returns):** main Agent dispatches one subtask to a sub-Agent (e.g. "collect," "analyze," "aggregate" three routes parallel), sub-Agent finishes returning result, **control always in main Agent's hands**, main Agent aggregates then decides next step. Doubao Work's "work squad" mainly this mode.
- **Handoff (responsibility transfer):** control whole baton handed to next owner, must pass `validate` to change owner. UDOS side's TransferBundle six fields (Goal/Context/Done/ToDo/Trace/Owner) + scope field designed for this mode (see lower paper comparison).

Chapter summary: intelligent agent/Agent/super Agent not synonyms, but autonomy tiers; nested containment+L0–L5 ladder can place Doubao six forms on one chart; Agent-as-Tool and Handoff are two essentially different multi-Agent collaboration contracts.

4 Six Forms × Nine Dimensions Matrix (Deliverable ③)

定位	目标用户	自主性	工具/执行环境	记忆层级	多Agent协作	权限模型	典型场景	入口
2	2	1	1	1	0	2	2	3
3	3	2	3	2	2	2	3	3
4	4	3	4	3	3	4	4	4
4	4	3	3	4	3	4	4	3
4	3	4	4	2	3	3	3	4
4	4	3	4	3	3	4	4	3

跃迁：①→③ SOP→目标驱动 | ③→④ 单人→团队独立身份 | ④/③→⑤ App内→OS级（MCP/A2A+GUI 催化）

完整 54 格文字版见正文原生大表；颜色深浅仅表示作者定性强度（0=弱→4=强），非实测评分

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Splitting the Doubao system's six externally usable forms along nine dimensions, obtaining the below 6×9=54-cell native master table. This table is the index for later SOP selection and PRD templates.

Form	Positioning	Target users	Autonomy	Tools and execution environment	Memory tier	Multi-Agent collaboration	Permission model
① C-end UGC intelligent agent (plaza)	Interactive role/Agent plaza, creatable and consumable	General C-end users, UGC creators	L1-L2	Name/avatar/persona/opening/voice/knowledge base/visibility; plugins and workflows published after Coze orchestration (config items to verify)	Single-role conversation memory	Itself single-role; multi-Agent orchestrated on Coze side	Creator self-sets visibility; plaza publishing must meet platform rules
② Coze orchestrated Agent	Developer-facing Agent building/orchestration platform	Developers, app builders	L2-L3	Persona/plugins/workflows/knowledge base/multi-Agent mode; database/cards/variables/triggers (part to verify)	Knowledge base+variables	Multi-Agent mode orchestration	Authorized per publishing channel; API independent callable
③ Doubao Work	Doubao's brand new Agent product and brand for productivity scenarios	Personal knowledge workers, professional office people	L3	Local+cloud PC dual mode; Windows virtual desktop, phone remote controlling computer; skill center 200+, connectors (DingTalk/Tianyancha etc. API Key authorization); browser cross-software operation; delivers docs/sheets/PPT/webpages/images/videos	Feishu account inherits enterprise knowledge and work context, creation sediments back to Feishu	Multi-specialty Agents form "work squad"; single task automatically splits multi-route sub-Agents parallel	Strictly inherits Feishu permission system; personal/enterprise data isolation device access quota control encryption, audit full-chain protection
④ Doubao Work Companion	Domestic first team Agent, originally Feishu aily renamed/upgraded	Enterprise/team all members, admin unified configuration	L3-L4	Within Feishu writing docs/bitable/schedules/meetings/approvals; Feishu CLI opens 767 function points	Team long-term memory (group chat history+business docs), one person teaches whole team can use	Can connect third-party/self-built/professional Agent teams (details to verify)	Independent identity/permissions/memory; enterprise unified configuration capability and usage; follow

Form	Positioning	Target users	Autonomy	Tools and execution environment	Memory tier	Multi-Agent collaboration	Permission model
							Feishu identity system
⑤ Doubao phone assistant	AI assistant cooperating with phone vendors at OS layer; no self-developed phone plan	Smartphone users (partner vendor terminals)	L5	UI-TARS closed-source substrate; system-level screenshot+event injection; MCP/A2A/SAEP; cross-App click/swipe	Optional memory, behavior personalization under user authorization	A2A interoperates with Meituan/Alipay etc. App built-in agents at bottom layer	System-level preinstalled app-level permissions; sensitive operations (publish/delete/logout/pay) need human review or secondary confirmation
⑥ Enterprise platform (AgentKit etc.)	Enterprise-facing Agent infrastructure and platforms, covering model to application full chain	Enterprise customers, enterprise developers	Per application defined	AgentKit/HiAgent 3.0/ArkClaw/AI Trust/Ark CLI; AI cloud-native architecture (GPU core)	Enterprise self-built knowledge base and data	Platform-level Agent orchestration	Enterprise tenant-level control; AI Trust security system

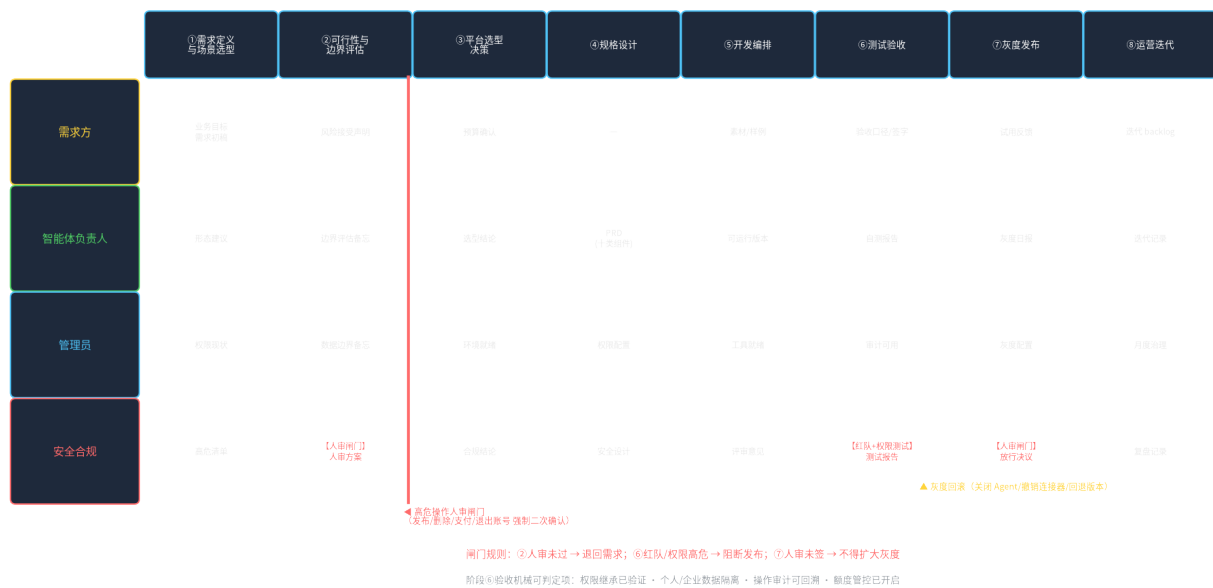
4.1 Three Leaps Between Forms

The spec sheet归纳 form evolution as three leaps (**not official explicit text**, author reverse-inferred from product release timeline):

- ① → ③ **from UGC to goal-driven**: UGC plaza's intelligent agent a "persona+plugins" conversation role; Doubao Work leaps to "planning→tools→execution→delivery" long-process autonomous Agent, can itself write proposals, do research, cross-software operate.
- ③ → ④ **from single person to team independent identity**: Doubao Work borrows user's personal Feishu permissions; Work Companion has its own independent identity/permissions/memory, all members share one "team memory," "one person teaches whole team can use."
- ④/③ → ⑤ **from in-App to OS-level**: Work Companion and Work still run in Apps; phone assistant leaps to OS-level cross-App operation, MCP/A2A + GUI event injection this leap's technical catalyst.
- ② → ① **development to consumption pipeline**: Coze orchestrated Agents published to Doubao plaza, a "developer→C-end user" distribution pipeline.

Chapter summary: six forms not six parallel products, but six tiers on the same autonomy ladder; three leaps catalyzed respectively by SOP capability, team identity, OS-level GUI three things.

5 Production SOP: Eight Stages × Four Roles Swimlane (Deliverable ④)



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The SOP solves: **an enterprise genuinely wanting to deploy an Agent for its team, starting from which step, who signs, at which step must stop and wait human review?** The spec sheet splits this process into eight stages, four roles, four explicit gates.

5.1 Four Roles and Eight Stages

Four roles: requester / Agent owner / admin / security compliance. **Eight stages:** ① requirement definition and scenario selection → ② feasibility and boundary assessment → ③ platform selection decision → ④ specification design → ⑤ development orchestration → ⑥ testing acceptance → ⑦ canary release → ⑧ operations iteration.

5.2 Per-Stage Input/Action/Output/Checklist

Stage	Input	Key actions	Output	Checklist / gate	Responsible role
1 Requirement definition and scenario selection	Business requirements, target users, expected effects	Judge task complexity, GUI needed, team collaboration, cost constraints; select among six forms	Scenario selection conclusion, user stories, requirement draft	Long-process autonomous? Cross-software? Team sharing needed? Sensitive operations involved?	Requester + Agent owner
2 Feasibility and boundary assessment	Scenario selection conclusion	Assess tool/connector availability, data permission scope, compliance risk, model and quota cost	Feasibility report, boundary statement (what to do/not do), risk acceptance statement	Data within permission scope? Sensitive operation list complete? Model and quota suffice?	Agent owner + security compliance
3 Platform selection decision	Feasibility report, decision tree	Per decision tree select: Doubao App/Coze/ Doubao Work/ Work Companion/ AgentKit	Platform selection decision table, budget confirmation	Zero code or high code? Enterprise API integration needed? Independent identity and all-member sharing needed?	Agent owner + admin
4 Specification design	Selection conclusion, PRD template	Per middle paper ten component field tables fill specifications item by item	One-page Agent PRD	Ten component categories all filled? Values evidence-grounded? To-verify items marked?	Agent owner
5 Development orchestration	PRD specification	Configure persona/prompts/tools/workflows/ knowledge base/ memory/ permissions	Runnable Agent version	Tool authorization complete? Knowledge base mounted? Permissions minimized?	Agent owner + admin
6 Testing acceptance	Runnable version	Functional/ boundary/ red-team adversarial / permission tests	Test report, acceptance conclusion, acceptance signature	[human review gate] sensitive operations mandatory human review/secondary confirmation? [red team] privilege escalation/ prompt injection/ inducement operations pass? [permission tests] "if employees cannot see, Agents cannot obtain" verified?	Agent owner + security compliance
7 Canary release	Accepted version	Small-scope directed co-	Canary observation	Canary scope controllable? Anomaly	Admin + Agent owner

Stage	Input	Key actions	Output	Checklist / gate	Responsible role
		creation/canary, monitor quota, error rate, anomalies	records, canary daily report, release decision	rollback ready? Quota capped? Human review gate unsigned cannot expand canary	
8 Operations iteration	Canary/full run data	Feedback closed loop, optimize prompts/tools/ knowledge base, update memory strategy	Iteration versions, operations review, review records	Feedback closed loop? Model upgrade regression tests? To-verify items against official originals?	Agent owner + requester

5.3 Four Explicit Gates (Must Draw on Swimlane Chart)

1. **High-risk operation human review gate:** publish/delete content, logout, pay etc. high-risk operations mandatory human takeover or secondary confirmation; stage ② human review gate not passed returns requirement, stage ⑦ human review gate unsigned cannot expand canary.
2. **Red-team adversarial testing:** privilege escalation, prompt injection, high-risk operation inducement; stage ⑥ red team high-risk items block release.
3. **Permission test checkpoint:** verify "**what employees cannot see, Agents cannot obtain**"; cross-tenant/cross-group/over-scope access cases all green.
4. **Canary rollback:** canary list/switches ready, observation period and on-duty person clear; rollback = close Agent / remove connectors / retreat version.

5.4 Stage ⑥ Acceptance Mechanically Decidable Items

Not relying on human feeling, four items writable as assertions:

- Permission inheritance verified (if employees cannot see, Agents cannot obtain);
- Personal/enterprise data isolation configured;
- Operation audit connected and traceable;
- Quota control enabled.

5.5 One-Page PRD Template

Field	Fill requirement
Name	External name / internal code
Positioning and target users	One sentence clearly for whom solving what problem
Selected form	Choose one of six forms (UGC/Coze/Doubao Work/Work Companion/phone assistant/enterprise platform)
Model and mode	Base model + run mode + reasoning mode
System prompt key points	Role, boundaries, prohibitions, output format
Tools and connectors and authorization	Which plugins/connectors, who provides API Key
Workflows and knowledge base	Multi-step orchestration nodes; which docs mounted
Memory strategy	Conversation memory / team long-term memory / behavior personalization / forgetting strategy
Multi-Agent collaboration	Single Agent / work squad / A2A interoperation
Permissions and high-risk operation list	Permission inheritance, human review/secondary confirmation items, enterprise unified configuration
Observable metrics	Operation audit, quota, success rate, red team results
Acceptance criteria	Stage ⑥ four mechanically decidable items + business cases
Canary plan and rollback	Canary scope, observation period, rollback operations
To-verify items	Which fields official not public, need follow-up

Chapter summary: the SOP's core not "eight-step process," but four gates—human review, red team, permission tests, canary rollback—separating the risk between "AI autonomous" and "humans not fallback"; stage ⑥'s four mechanically decidable items make acceptance not rely on gut.

6 Ten Component Field Specifications (Deliverable ⑤)

These ten tables from the spec sheet middle paper, the complete list of "which fields configuring one Agent must fill." **Fields only added not deleted, "to verify" retained as-is.**

6.1 Component One: Identity and Persona

Field name	Description	Required	Values or examples	Configuration platform
Name	External display name	yes	e.g. "industry research assistant," "announcement interpretation assistant"	App Agent / Coze
Avatar	Role visual identifier	no	Upload image	App Agent / Coze
Persona and role setting	Identity, tone, behavior rules	yes	"You are a rigorous financial report analyst, only answer based on given materials"	Coze / Doubao Work
Opening line	First conversation guidance	no	"I can help you interpret listed company announcements"	App Agent (to verify)
Voice	Voice broadcast timbre	no	Timbre selection (plaza config items to verify)	App Agent
Visibility	Public/team/private/specified scope	yes	Public plaza / private / specified scope	App Agent / Coze / Work Companion
Independent identity (team level)	Independent Agent identity区别于 "borrowing user permissions"	Team level required	Doubao Work Companion independent identity	Work Companion (Feishu admin configuration)

6.2 Component Two: Model and Mode Selection

Field name	Description	Required	Values or examples	Configuration platform
Model version	Selected base model	yes	Doubao 2.1 Pro / 2.1 Turbo / 2.0 (Pro/Lite/Mini) /1.5 deep thinking	Doubao Work / Coze / AgentKit
Run mode	Interaction / task mode	yes	Expert mode / office task (Turbo) / conversation	Doubao Work
Reasoning mode	Deep thinking or not	no	Deep thinking for complex reasoning	Doubao Work / Coze
Subscription tier	Pro version ladder	no	Standard 68 / enhanced 200 / premium 500 yuan/month (standard quota over 5× free; enhanced=standard 4×, premium=standard 10×)	Doubao Work (C-end subscription)
API unit price	Enterprise call billing	no	2.1 Pro input 6 yuan / output 30 yuan per million Tokens; Turbo about half	AgentKit / Ark
GUI substrate	Selected when needing to operate system interfaces	as needed	UI-TARS closed-source version (Mobile Use optimized)	Phone assistant (vendor integration)
Context length/ concurrency/ quota	Long context and concurrency cap	no	Official not public details	—

6.3 Component Three: System Instructions • Prompts

Field name	Description	Required	Values or examples	Configuration platform
Role setting/ system prompt	Identity and professional boundaries	yes	Containing goals, boundaries, prohibitions, output format	Coze / Doubao Work / AgentKit
Behavior rules	Must/must not do what	yes	State uncertainty, not fabricate numbers; not exceed user permission scope	Coze / Doubao Work
Tone style	Formal/affable/ professional	no	Restrained, objective, point-wise output	Coze
Refusal and security boundaries	Sensitive/over-privilege problem handling	yes	Sensitive operations to human review	Work Companion / AgentKit
Output format	Structured delivery requirements	no	Docs/sheets/PPT/webpages, support "point where change where"	Doubao Work
Custom skills (prompt derived)	Conversation-generated or uploaded skills	no	Performance analysis, financial reports, de-AI-flavor etc.	Doubao Work (skill center)

6.4 Component Four: Tools and Connectors

Field name	Description	Required	Values or examples	Configuration platform
Plugins	Extending tool call capability	no	Search, code, image, calculator etc.	Coze (Plugin)
Connectors	Connect external apps, need API Key authorization	as needed	DingTalk, Tianyancha etc.; user provides Key authorization	Doubao Work (connectors)
Browser/cross-software operation	Web and local software operation	as needed	Fill forms, compare materials, use professional software to process data	Doubao Work
Execution environment	Runtime environment selection	yes	Local computer / cloud PC / Windows virtual desktop / phone remote	Doubao Work
MCP tool integration	Application-side opened MCP services	as needed	Only under premise application side actively opens and authorizes	Phone assistant (second generation)
Skill/connector scale	Listed quantity reference	reference	As of 2026-08-21 over 200	Doubao Work
Tool governance/ authorization method	Review/billing/interface specs	no/ required	Official not public details; minimal authorization per tool requirements	— / AgentKit

6.5 Component Five: Workflows and Skills

Field name	Description	Required	Values or examples	Configuration platform
Workflow	Multi-step task orchestration	as needed	Collect→analyze→aggregate report (nodes串联)	Coze (Workflow) / Doubao Work
Built-in skills	Skill center prebuilt capabilities	no	PPT, financial reports, announcement interpretation, short drama creation, ecommerce selection, medical literature search, code review etc. (200+)	Doubao Work (skill center)
Custom skills	Upload or conversation-generated skills	no	Customized per team scenarios	Doubao Work
Triggers	Scheduled/event triggered	no	Daily stock sentiment briefing; scheduled group message organization (official mechanism to verify)	Coze / Work Companion
Variables/cards	Runtime parameters and interaction cards	no	Variable passing / card interaction (official mechanism to verify)	Coze

6.6 Component Six: Knowledge Base and Data

Field name	Description	Required	Values or examples	Configuration platform
Knowledge base	Connect docs as knowledge sources	as needed	Enterprise/personal doc upload and retrieval	Coze / Doubao Work / Work Companion
Database	Structured data read/write	no	Structured storage (official mechanism to verify)	Coze
Feishu enterprise knowledge inheritance	In-permission-scope context	Team level	Group chat records, docs, meeting minutes, schedules; creation sediments back to Feishu	Doubao Work / Work Companion
Data isolation	Personal and enterprise data boundary	yes	Personal data and enterprise data isolated (official caliber)	Doubao Work
Data retention strategy	Cloud PC/session data retention	no	Official not public details	—

6.7 Component Seven: Memory

Field name	Description	Required	Values or examples	Configuration platform
Conversation memory	Single-role/single-session context	yes	Single-role conversation memory	App Agent
Team long-term memory	Team-level sediment	Team level	Group chat history + business docs; one person teaches whole team can use	Work Companion / Doubao Work
Behavior personalization memory	Common behavior under authorization	no	Personalized through common behavior under user authorization	Phone assistant
Memory cleanup/forgetting strategy	Forgetting/clearing policy	no	Enterprise control	Work Companion / AgentKit
Cross-Agent long-term memory	Shared memory across multiple Agents	no	Not C-end UGC main capability	—

6.8 Component Eight: Task Planning and Multi-Agent Collaboration

Field name	Description	Required	Values or examples	Configuration platform
Task decomposition	Autonomously decompose per goal	Long process required	Data collect/analyze/aggregate three-route parallel sub-Agents	Doubao Work
Work squad	Multi-specialty Agent division	as needed	Multiple specialty Agents team formation collaboration	Doubao Work
Team formation/multi-Agent mode	Connect external/self-built Agents	as needed	Research-output-check, main Agent aggregation (official explicit text to verify)	Coze / Work Companion
A2A interoperation	Cross-application Agent bottom-layer interoperation	as needed	Interoperate with Meituan/Alipay etc. App built-in agents	Phone assistant (second generation)
Long-time running	Long process/scheduled tasks	no	Cloud PC long-time running; scheduled long-term tasks	Doubao Work

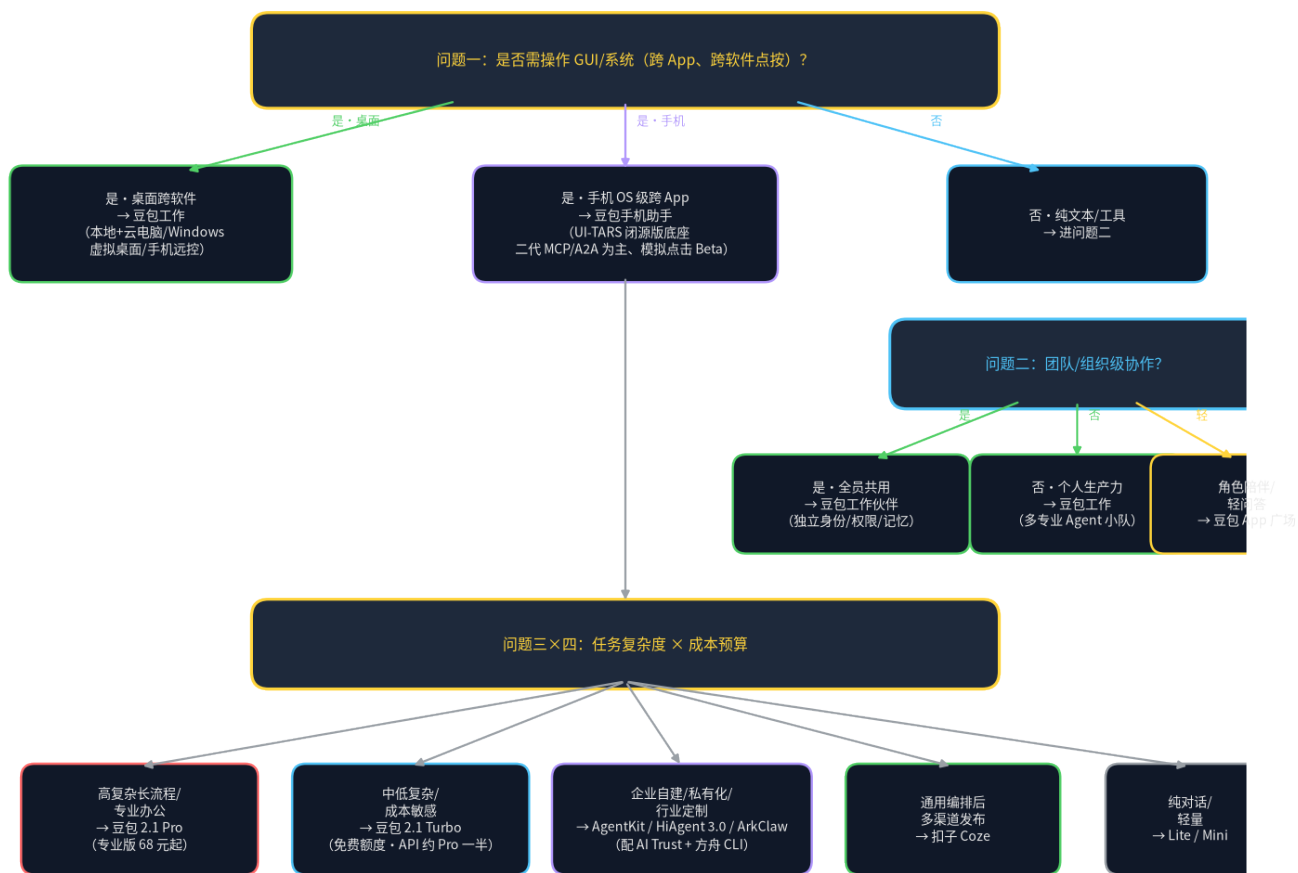
6.9 Component Nine: Permissions • Security • Compliance

Field name	Description	Required	Values or examples	Configuration platform
Permission inheritance	Follow existing identity system	yes	Strictly inherits Feishu permissions; content employees cannot see Agents cannot obtain	Doubao Work / Work Companion
Device and access control	End-to-end protection	yes	Device access, permission settings, quota control, data encryption, operation audit	Doubao Work (official caliber)
Human review/ secondary confirmation gate	High-risk operation human takeover	High risk required	Publish/delete content, logout, pay etc. need human takeover or secondary confirmation	Phone assistant / all forms
Enterprise unified configuration	Team-level capability and usage	Team level required	Admin unified configuration permissions, capability, usage	Work Companion
Independent identity	Team Agent independent identity	optional	Enterprise unified configuration independent identity/permissions/ memory	Work Companion
Security product system	Enterprise Agent security	Enterprise level	AI Trust enterprise Agent security system	AgentKit / ArkClaw
Qualifications and white papers	Compliance certification	no	Domestic first batch office Agent capability and cloud benchmark dual certification; AI privacy and security white paper	Doubao Work / phone assistant
SAEP declaration	Third-party app operation boundaries	System level	App autonomously declares allowing/refusing/limiting AI automated operations (30-day notice)	Phone assistant (ecosystem side)

6.10 Component Ten: Observability and Evaluation

Field name	Description	Required	Values or examples	Configuration platform
Operation audit	Full-chain operation trace	Enterprise level required	Operation audit (field details to verify)	Work Companion / AgentKit
Quota control	Usage and quotas	yes	Quota control; pro version per tier multiples	Doubao Work / AgentKit
Call success rate	Tool/interface stability	no	Feishu CLI call success rate 78% → 95%	Feishu CLI (baseline) / Work Companion
Red-team adversarial testing	Pre-launch adversarial evaluation	High risk required	Privilege escalation/injection/high-risk operation inducement tests	AgentKit / self-built
Permission tests	Over-privilege access verification	Enterprise level required	Verify "if employees cannot see, Agents cannot obtain"	Work Companion / AgentKit
Canary strategy	Release scope	optional	Directed co-creation/small-scope canary (Work Companion as of 2026-09-18 still in directed co-creation stage)	Work Companion / AgentKit

6.11 Model and Mode Selection Decision Tree



边界：Turbo/Pro 并发·算力配额·上下文长度差异官方未公开明细，选型以官方计费页为准

决策树为公开资料归纳·示意·2026-09-21

Decision tree four questions:

- 1. Need to operate GUI/system?** Cross-desktop/software → Doubao Work; phone OS-level cross-App → Doubao phone assistant; pure text/tools → to question two.
- 2. Team/organization-level collaboration?** All members share → Doubao Work Companion; personal productivity → Doubao Work; role companionship/light Q&A → Doubao App Agent plaza.
- 3. Task complexity × cost:** high complexity long process professional office → 2.1 Pro (68 yuan starting); medium-low complexity/cost sensitive → 2.1 Turbo (free quota, API about half Pro); enterprise self-built/privatization/industry customization → AgentKit/HiAgent 3.0/ArkClaw; general orchestration multi-channel publishing → Coze; pure conversation/lightweight → Lite/Mini.

Boundary note: Turbo/Pro concurrency, compute quota, context length differences official not public details, selection per official billing page.

Chapter summary: ten component categories ≈ one Agent's "configuration surface"; filling these ten tables, the one-page PRD written; decision tree four questions make "which product/which model" from gut to askable-answerable.

7 Tech Stack Timeline and UI-TARS End-to-End Principle

7.1 Tech Stack Version Timeline (2024-05 to 2026-09)

Date	Event	Three-state
2024-05-15	Doubao large model (Skylark) first release, 9 models	official
2025-01-22	Doubao-1.5-pro released; UI-TARS paper arXiv:2501.12326 submitted	official/paper
2025-04-17	Doubao 1.5 deep thinking + UI-TARS-1.5 open source; OS Agent plan released	official
2025-05-13	Seedance 1.0 lite video generation	official
2025-12-01	First-generation nubia M153 sale, 3499 yuan, stocked 30,000 first batch sold out	official/media
2025-12-18	Doubao 1.8 released, daily tokens broke 5 trillion	official
2026-02-14	Doubao 2.0 (Pro/Lite/Mini/Code) released	official
2026-03-09	ArkClaw cloud SaaS OpenClaw launched (encyclopedia caliber)	official/media
2026-06-24	Doubao 2.1 Pro/Turbo + AgentKit upgrade + HiAgent 3.0 + AI Trust + Ark CLI; Doubao pro version online (68/200/500)	official
2026-07-30	Feishu product team+Doubao product team merged into new Doubao product team (Zhao Qi responsible)	official/media
2026-08-14	Feishu aily renamed Doubao Work Companion (official help center)	official
2026-08-21	Doubao Work skill center listed over 200 skills and connectors	official/media
2026-08-24	Coze, TRAE Work overall merged into Doubao system; TRAE IDE/CLI retained programming line	official response/media says
2026-08-25	Doubao Work independent client formally released	official
2026-09-01	Doubao Work Companion adds "proactive work" and bills	official
2026-09-14	Doubao phone assistant consumer version released; SAEP enters 30-day notice period	media says (Doubao launched)
2026-09-15	Feishu Future Infinite conference: Work Companion publicly released, Feishu 8.0 released, Xie Xin "Feishu also serves Agents"	official/press conference retelling

7.2 UI-TARS GUI Agent End-to-End Principle

UI-TARS is Doubao phone assistant's GUI operation substrate. Its end-to-end closed loop five steps:

1. **Screenshot perception:** system-level screenshot obtains current screen pixels;
2. **Reasoning decomposition:** model splits "help me order a coffee on Meituan" into "open Meituan→search coffee shop→select item→order→pay";
3. **Element localization:** on screenshot localize element coordinates to click/swipe;
4. **Event injection:** via Android `WindowManagerService` / `inject_events` inject click/swipe events;
5. **Reflective correction:** after operation screenshot again, judge whether expected interface reached, if not retry.

Key boundary: `inject_events` type system-level event injection capability **only open to vendor preinstalled apps**—also why Doubao explicitly "no self-developed phone plan," choosing to cooperate with

Nubia etc. vendors at OS layer. Ordinary third-party Apps cannot obtain this permission, also the problem SAEP declarative boundaries solve: application side itself declares which operations allowed, which refused.

UI-TARS 2.0 direction (media says/to verify): points out pure GUI screenshot perception limits (slow, fragile, unstable), starts connecting external file systems and sandbox platforms via SDK, taking "half GUI half API" route.

Chapter summary: in the two-plus year timeline, Doubao went from "producing models" to "producing OS Agents"; UI-TARS's five-step closed loop is the GUI Agent general paradigm, but system-level event injection permission boundaries determine it must take vendor cooperation route, not make its own phone.

8 Glossary

Term	This paper's usage
Intelligent agent (Agent)	Doubao official uniformly uses "Agent"; Chinese "intelligent agent" its translation
Super Agent	Not official product name , media umbrella for "system-level OS Agent / multi-Agent organization"
OS Agent	Operating system-level AI assistant, can cross-App screenshot+inject events
MCP	Model Context Protocol
A2A	Agent-to-Agent
SAEP	Screen Automation Execution Protocol (launched by Doubao)
UGC	User-generated content, here Doubao App Agent plaza user-built agents
RAG	Retrieval-augmented generation, knowledge base mounting general practice
Human review gate	High-risk operations mandatory human takeover or secondary confirmation
Red team tests	Actively attack tested Agent with privilege escalation/prompt injection/inducement
Agent-as-Tool	Sub-Agent finishes work returning control to main Agent
Handoff	Responsibility whole baton transferred to next owner, validate passed to change
Three states	[official statement][media says][to verify] evidence grading

9 Failure Boundaries and Falsifiability

9.1 This Paper's Failure Boundaries

1. **Timestamp:** this paper based on public materials as of 2026-09-21. The Doubao system iterates fast monthly (2026-06-24, 07-30, 08-24, 08-25, 09-14, 09-15 all big nodes), this paper's conclusions may stale after 1–2 months.
2. **Three states not upgraded:** all numbers and dates marked [media says][to verify], this paper not writing them as official facts; spec sheet self-listed 14 "to verify" retained as-is (including "super Agent" product name, Turbo/Pro concurrency and context length, skill governance details, cloud PC specs and data retention, Work Companion multi-Agent division mechanism, permission model

details, UI-TARS 2.0 date, Tan-Dai dichotomy original words, Coze merge original URL, Volcano Engine docs, Ark annual Token call volume, UGC plaza config items, Coze complete component matrix, Work Companion full timeline).

3. **Not constituting procurement advice:** this paper only technical anatomy, not recommending anyone's commercial decisions.
4. **Qualitative judgments overturnable:** Figure 3 heat matrix, Figure 2 autonomy landings, inter-form leap descriptions, all author's qualitative induction based on public materials, not official scores, not measured.

9.2 Five Falsifiable Predictions

1. **If Doubao in 2026 Q4 releases Work Companion multi-Agent division white paper** (explicitly writing how "research-output-check" splits, how main Agent aggregates), then this paper Section 6.8 "official explicit text to verify" should upgrade to [official statement].
2. **If Doubao publicly releases Turbo/Pro concurrency and context length**, then this paper Section 6.2 "official not public details" should be replaced with specific numbers.
3. **If after SAEP 30-day notice period ends, head Apps (Meituan/Alipay/WeChat) publicly declare integrating or refusing SAEP**, then this paper Section 2.2 "actual integration rate to observe" should update to fact.
4. **If Doubao phone assistant in some 2027 release removes the "tech preview" label, publishes cross-App task success rate**, then this paper Figure 2 "L5 not achieved" should reassess.
5. **If Volcano Engine publicly releases ArkClaw 3-09 launch official version records** (currently first version Ark-26.3.30/03-31), then this paper ArkClaw date three-state should upgrade from "official existence+media says date."

10 References (Excerpt)

The following representative entries of this paper's fact sources, complete ledger see scratch/fact ledger_v773.md.

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Upper paper ends. The lower paper "The Doubao System and the UDOS Reasoning Engine: A Technical Comparison of Industrial Productization and a Falsifiable Governance Substrate" will horizontally compare this paper's Doubao status with UDOS's BFT-lite / TransferBundle / quality saturation law.